

Data Analyst Internship Record Report

Student Details

Name: Pallavi Singh

Project Title: Uber Demand-Supply Gap Analysis

Tools Used: Excel, SQL, Python, Power BI, GitHub

Problem Statement

Uber faces a demand-supply gap where customer ride requests are not always fulfilled due to driver shortages, cancellations, and unavailability of cars. This affects customer satisfaction and business performance.

Objective

The objective of this project is to analyze Uber ride request data and identify:

- Peak demand hours
 - Driver availability issues
 - Trip cancellation patterns
 - Demand-supply gaps
 - Opportunities for business improvement
-

Excel Analysis

AutoSave 12-07-2016 09:17:00 Uber Analysis project dashboard - Saved to this PC

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query

Clipboard: Cut, Copy, Paste, Format Painter

Font: Calibri, 11, A, B, I, U, Color, Background Color

Alignment: Wrap Text, Merge & Center

Number: Custom, %, 0.00, 00

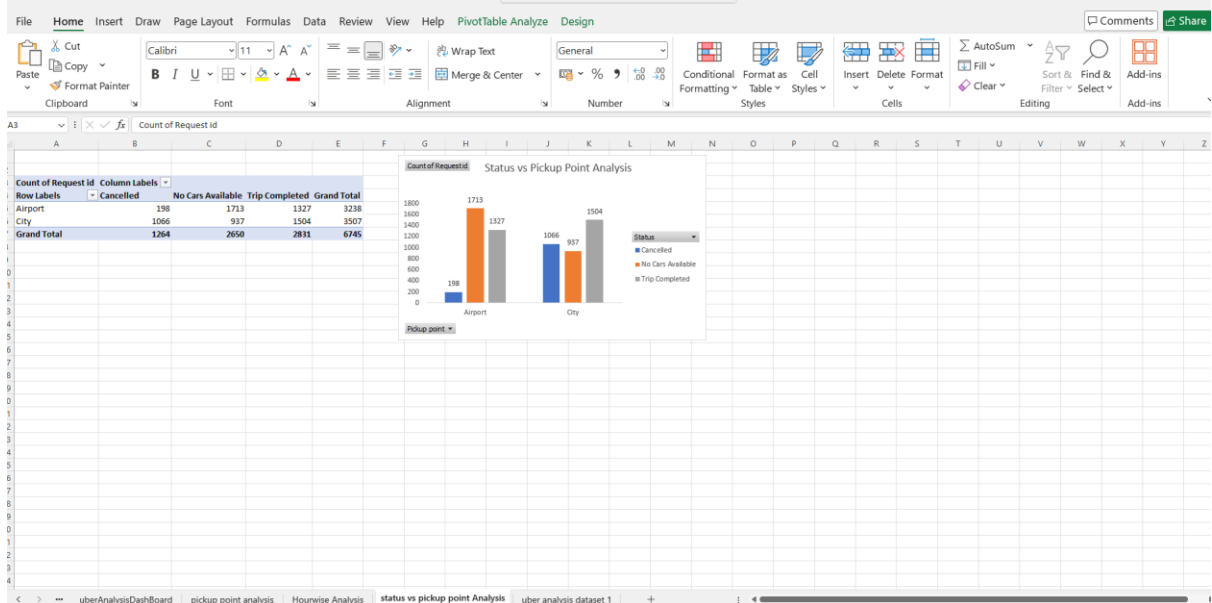
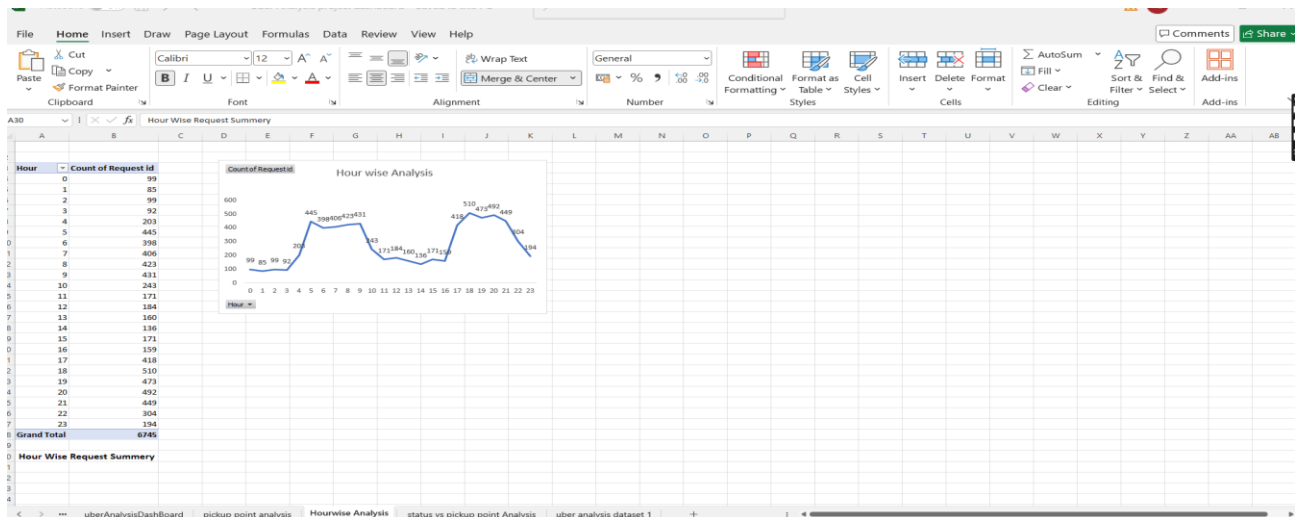
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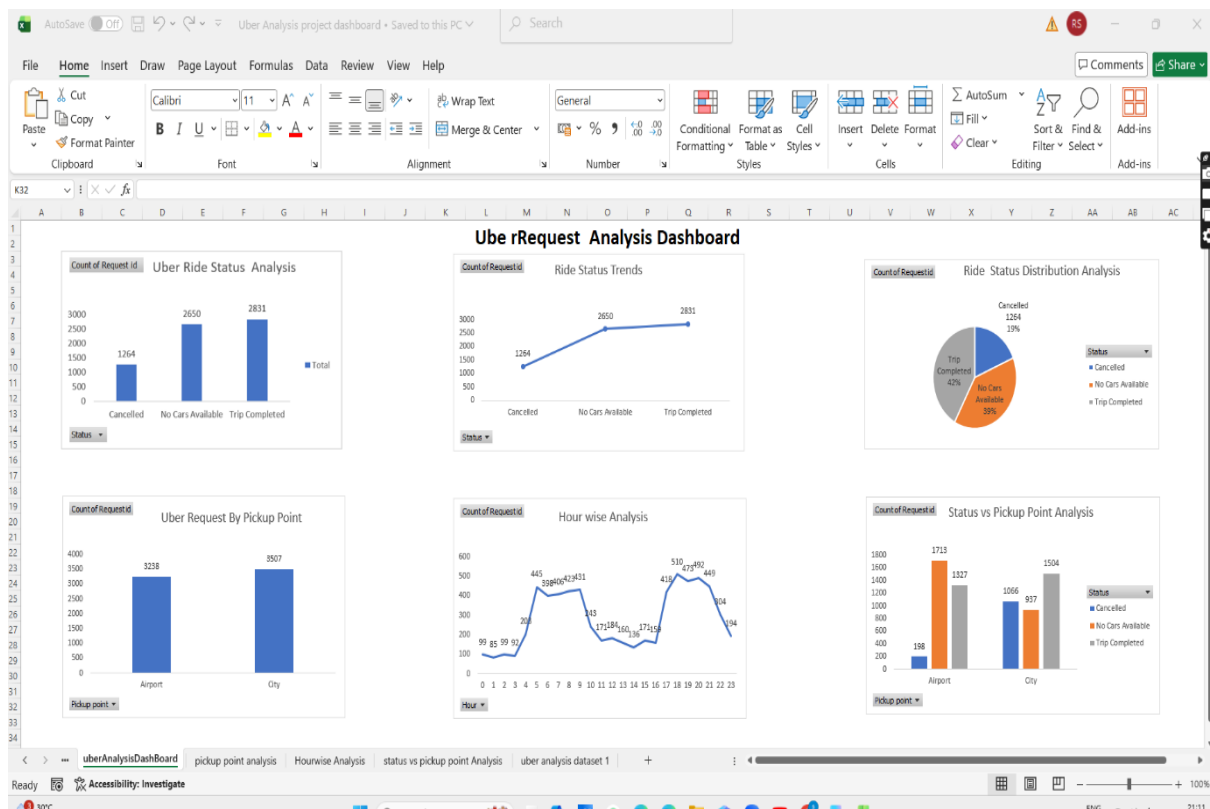
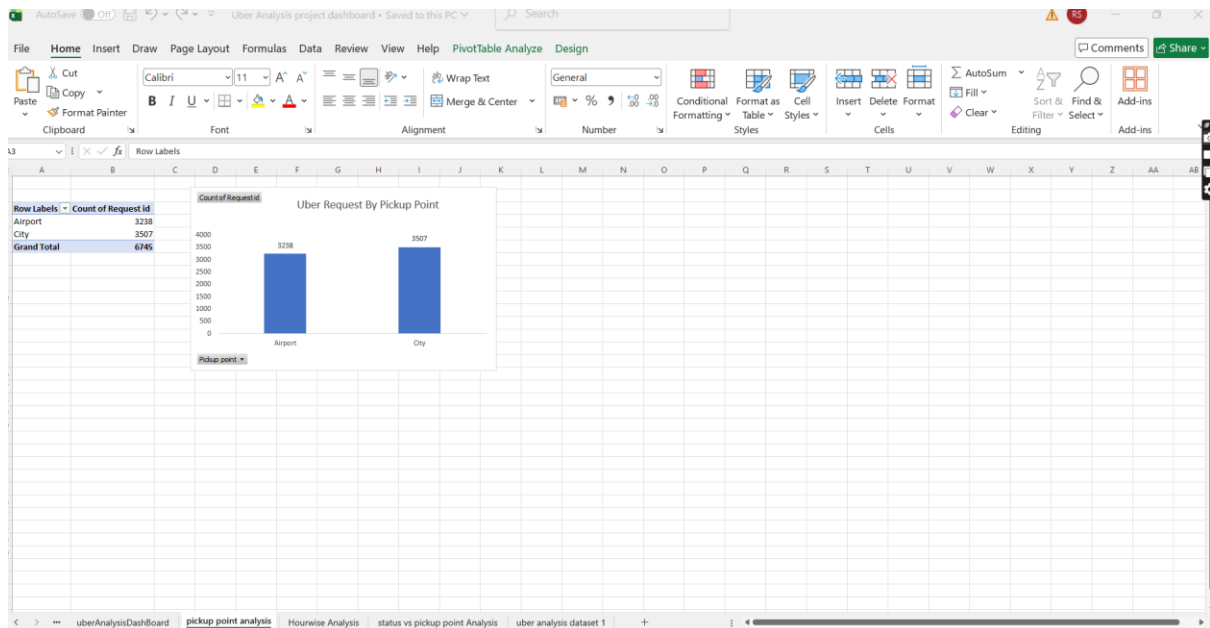
AutoSum, Fill, Clear, Sort & Filter, Find & Select, Add-ins

Comments, Share

Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp	Hour
619	Airport	1	Trip Completed	11-07-2016 11:51	11-07-2016 13:00	11
867	Airport	1	Trip Completed	11-07-2016 17:57	11-07-2016 18:47	17
1807	City	1	Trip Completed	12-07-2016 09:17	12-07-2016 09:58	9
2532	Airport	1	Trip Completed	12-07-2016 21:08	12-07-2016 22:03	21
3112	City	1	Trip Completed	13-07-2016 08:33	13-07-2016 09:25	8
3879	Airport	1	Trip Completed	13-07-2016 21:57	13-07-2016 22:28	21
4270	Airport	1	Trip Completed	14-07-2016 06:13	14-07-2016 07:13	6
5510	Airport	1	Trip Completed	15-07-2016 05:11	15-07-2016 06:07	5
6248	City	1	Trip Completed	15-07-2016 17:57	15-07-2016 18:50	17
287	City	2	Trip Completed	11-07-2016 06:46	11-07-2016 07:25	6
1467	Airport	2	Trip Completed	12-07-2016 05:08	12-07-2016 06:02	5
1383	City	2	Trip Completed	13-07-2016 13:30	13-07-2016 13:57	13
2784	Airport	2	Trip Completed	13-07-2016 04:49	13-07-2016 05:23	4
3075	City	2	Trip Completed	13-07-2016 08:02	13-07-2016 09:16	8
3979	City	2	Trip Completed	13-07-2016 14:23	13-07-2016 15:35	14
3482	Airport	2	Trip Completed	13-07-2016 17:23	13-07-2016 18:20	17
4652	City	2	Trip Completed	14-07-2016 12:01	14-07-2016 12:36	12
5135	Airport	2	Trip Completed	14-07-2016 22:24	14-07-2016 23:18	22
535	Airport	3	Trip Completed	11-07-2016 10:00	11-07-2016 10:31	10
960	Airport	3	Trip Completed	11-07-2016 18:45	11-07-2016 19:23	18
1934	Airport	3	Trip Completed	12-07-2016 11:17	12-07-2016 12:23	11
2083	Airport	3	Trip Completed	12-07-2016 15:40	12-07-2016 16:40	15
2311	Airport	3	Trip Completed	12-07-2016 18:00	12-07-2016 18:28	18
3096	Airport	3	Trip Completed	13-07-2016 08:17	13-07-2016 09:22	8
3881	Airport	3	Trip Completed	13-07-2016 21:54	13-07-2016 22:51	21
5254	City	3	Trip Completed	14-07-2016 21:23	14-07-2016 22:25	21
5434	City	3	Trip Completed	15-07-2016 02:41	15-07-2016 03:24	2
5916	City	1	Trip Completed	15-07-2016 10:00	15-07-2016 10:53	10
699	City	4	Trip Completed	11-07-2016 13:08	11-07-2016 13:49	13
1567	Airport	4	Trip Completed	12-07-2016 06:21	12-07-2016 07:10	6
1826	City	4	Trip Completed	12-07-2016 09:27	12-07-2016 10:27	9
2102	City	4	Trip Completed	12-07-2016 16:09	12-07-2016 17:01	16
2847	Airport	4	Trip Completed	13-07-2016 05:34	13-07-2016 06:09	5
3722	Airport	4	Trip Completed	13-07-2016 19:57	13-07-2016 20:32	19

uberAnalysisDashboard | pickup point analysis | Hourwise Analysis | status vs pickup point Analysis | uber analysis dataset 1





1. Uber Ride Status Analysis (Clustered Column Chart)

Why Used:

To compare the number of requests by ride status.

Observation:

Trip Completed requests are highest (2831), followed by No Cars Available (2650), and Cancelled requests (1264).

Business Problem:

A large number of requests could not be fulfilled due to driver shortages and cancellations.

Recommendation:

Increase driver availability and reduce cancellations through better allocation.

2. Ride Status Trends (Line Chart)**Why Used:**

To show the trend across different ride statuses.

Observation:

The number of completed trips is higher than cancelled trips, but No Cars Available remains significantly high.

Business Problem:

Demand is exceeding available driver supply during certain periods.

Recommendation:

Improve supply planning and demand forecasting.

3. Ride Status Distribution Analysis (Pie Chart)**Why Used:**

To show the percentage share of each ride status.

Observation:

Trip Completed contributes the largest share, while No Cars Available also represents a substantial portion of requests.

Business Problem:

Many customers are unable to get rides because cars are unavailable.

Recommendation:

Deploy more drivers in high-demand locations.

4. Uber Request by Pickup Point (Clustered Column Chart)**Why Used:**

To compare ride requests from Airport and City locations.

Observation:

City generated 3507 requests, while Airport generated 3238 requests.

Business Problem:

Demand varies by pickup location, creating location-specific supply gaps.

Recommendation:

Allocate drivers based on location-wise demand patterns.

5. Hour Wise Analysis (Line Chart)**Why Used:**

To identify peak demand hours throughout the day.

Observation:

Demand peaks during early morning (5–9 AM) and evening (5–9 PM).

Business Problem:

Driver shortages during peak hours increase cancellations and unavailable rides.

Recommendation:

Provide incentives to drivers during peak demand hours.

6. Status vs Pickup Point Analysis (Clustered Column Chart)

Why Used:

To analyze ride status across Airport and City locations.

Observation:

Airport has a higher number of "No Cars Available" requests, while the City has more completed trips.

Business Problem:

The Airport experiences a larger demand-supply gap than the City.

Recommendation:

Deploy additional drivers near airports and transportation hubs.

Dashboard Conclusion

Overall Finding:

The analysis shows that Uber faces significant demand-supply challenges, especially during peak hours and at airport pickup locations. The major reasons for unsuccessful requests are driver unavailability and ride cancellations. Increasing driver availability, improving allocation strategies, and monitoring demand patterns can help improve trip completion rates and customer satisfaction.

SQL Analysis

MySQL Workbench interface showing a SQL query and its output. The query is executed against the `uber_project` database, specifically the `uber_requests` table.

```
1 • SELECT * FROM uber_project.uber_requests;
2
3 • USE uber_project;
4
5 • SELECT COUNT(*) AS Total_Rows
6   FROM uber_requests;
7
8 • SELECT MIN(request_id), MAX(request_id)
9   FROM uber_requests;
10
11 • SELECT COUNT(*) AS Total_Rows
12   FROM uber_project.uber_requests;
13
14 • SELECT COUNT(*) FROM uber_requests;
15 • SELECT COUNT(DISTINCT request_id)
16   FROM uber_requests;
17
18
19 # Requests by Pickup Point and Status
20 # How many requests were received from each pickup location?
```

Output:

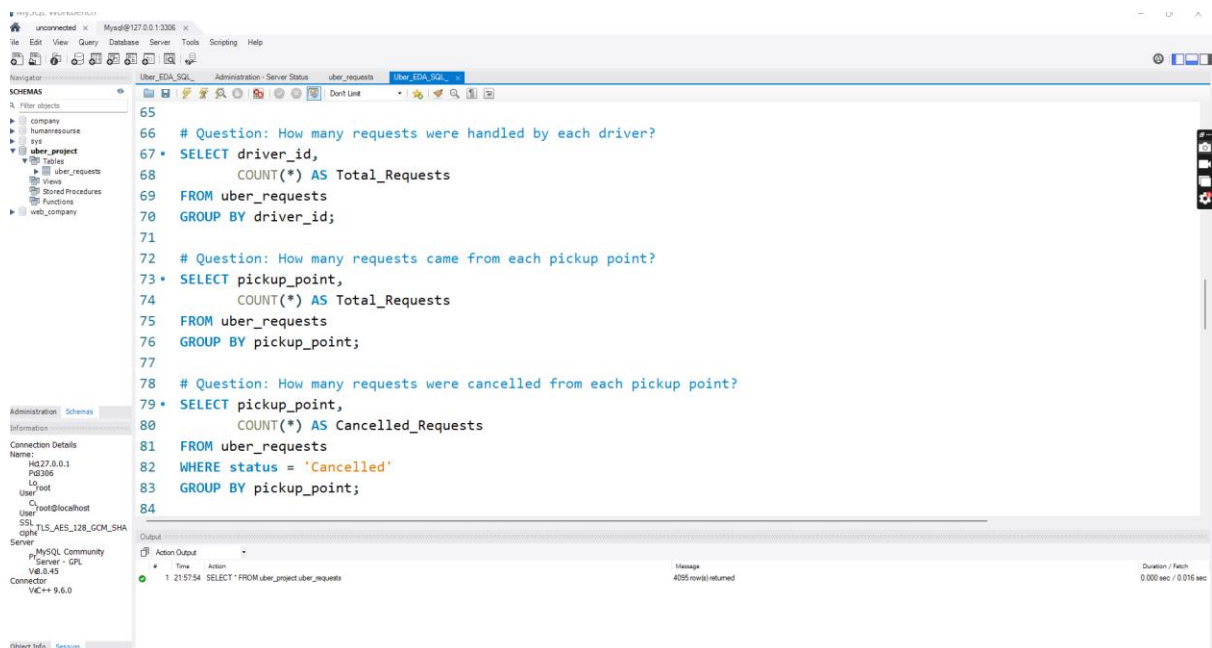
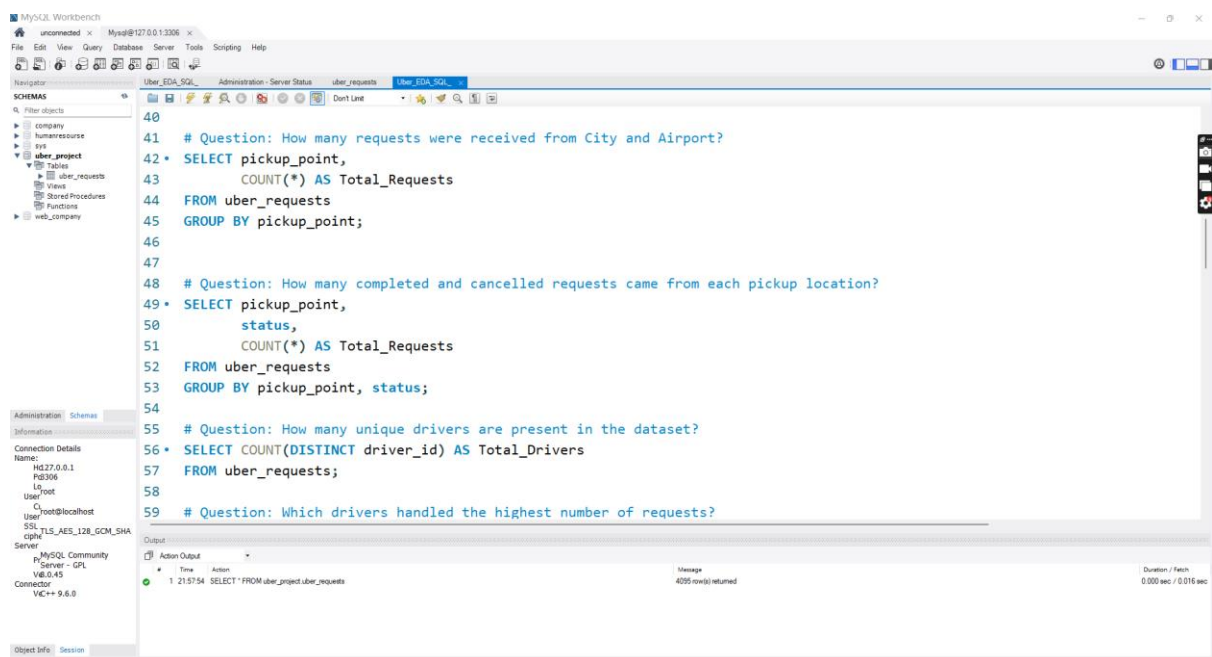
#	Time	Action	Message	Duration / Fetch
1	21:57:54	SELECT * FROM uber_project.uber_requests	4095 row(s) returned	0.000 sec / 0.016 sec

MySQL Workbench interface showing a SQL query and its output. The query is executed against the `uber_project` database, specifically the `uber_requests` table.

```
19 # Requests by Pickup Point and Status
20 # How many requests were received from each pickup location?
21 • SELECT pickup_point,
22         status,
23         COUNT(*) AS Total_Requests
24   FROM uber_requests
25  GROUP BY pickup_point, status;
26
27 # To determine the total number of requests from each pickup location
28 • SELECT pickup_point,
29         COUNT(*) AS Total_Requests
30   FROM uber_requests
31  GROUP BY pickup_point;
32
33
34 # Question: How many requests were completed and cancelled?
35 • SELECT status,
36         COUNT(*) AS Total_Requests
37   FROM uber_requests
38  GROUP BY status;
```

Output:

#	Time	Action	Message	Duration / Fetch
1	21:57:54	SELECT * FROM uber_project.uber_requests	4095 row(s) returned	0.000 sec / 0.016 sec



Work Performed

- Data Extraction
- Aggregation Functions
- GROUP BY Analysis
- Sorting and Filtering

Outcome

SQL queries identified peak demand periods, cancellation counts, and trip completion statistics.

Power BI Dashboard

Features

- Interactive Filters
- KPI Cards
- Trend Analysis
- Demand vs Supply Visualization

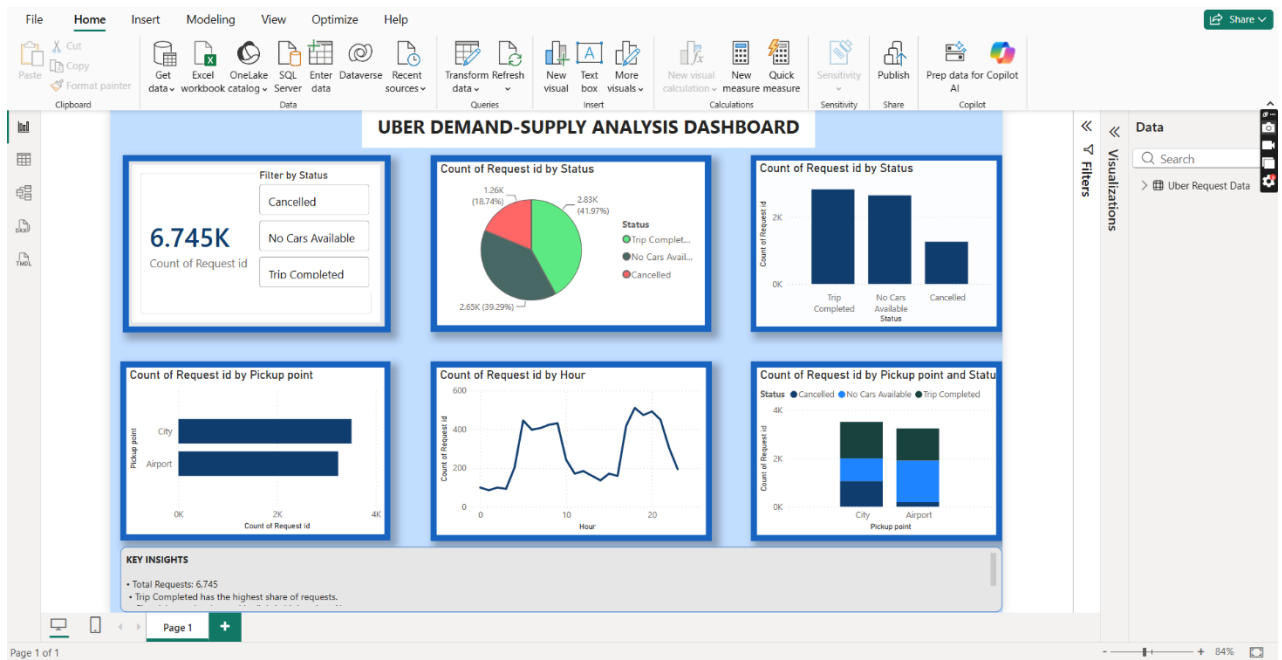
Outcome

Power BI provided an interactive view of business performance and demand patterns.

Solution and Recommendations

1. Increase driver availability during peak hours.
 2. Provide incentives for drivers near airports.
 3. Improve driver allocation using demand forecasting.
 4. Reduce cancellations through better operational planning.
 5. Monitor demand-supply gaps continuously.
-

Power BI Dashboard Analysis



1. KPI Card – Total Requests

Purpose

The KPI Card is used to display the total number of ride requests in a simple and easy-to-understand format.

Observation

- Total ride requests recorded: **6,745**
- The dataset contains a significant volume of customer ride requests.

Business Insight

This KPI provides a quick overview of overall business activity and demand.

2. Status Filter (Slicer)

Purpose

The slicer is used to filter dashboard visuals based on ride status.

Observation

- Users can analyze Cancelled, No Cars Available, and Trip Completed requests separately.

Business Insight

Interactive filtering helps stakeholders focus on specific operational issues.

3. Pie Chart – Request Status Distribution

Purpose

The pie chart is used to show the percentage distribution of ride requests by status.

Observation

- Trip Completed represents the largest share of requests.
- A considerable percentage of requests fall under "No Cars Available."
- Cancelled trips also contribute significantly to overall requests.

Business Problem Identified

The presence of "No Cars Available" requests indicates a demand-supply imbalance.

4. Clustered Column Chart – Request Count by Status

Purpose

This chart is used to compare the number of requests across different ride statuses.

Observation

- Trip Completed has the highest number of requests.
- No Cars Available is the second highest category.
- Cancelled requests remain a noticeable portion of total requests.

Business Problem Identified

Driver availability remains a challenge during certain periods.

5. Bar Chart – Request Count by Pickup Point

Purpose

This chart compares ride requests between Airport and City pickup locations.

Observation

- The City pickup point generated slightly more requests than the Airport.
- Demand is high at both pickup locations.

Business Insight

Understanding location-wise demand helps optimize driver allocation.

6. Line Chart – Request Count by Hour

Purpose

The line chart is used to identify hourly demand patterns.

Observation

- Demand increases significantly during morning hours.
- Another peak is observed during evening hours.

- Mid-day demand remains relatively lower.

Business Problem Identified

High demand during peak hours may result in driver shortages and service delays.

7. Stacked Column Chart – Pickup Point and Status Analysis

Purpose

This chart analyzes ride status distribution across different pickup locations.

Observation

- Airport locations show a higher number of "No Cars Available" requests.
- The City location records a higher number of completed trips.

Business Problem Identified

Airport areas experience greater supply shortages compared to City locations.

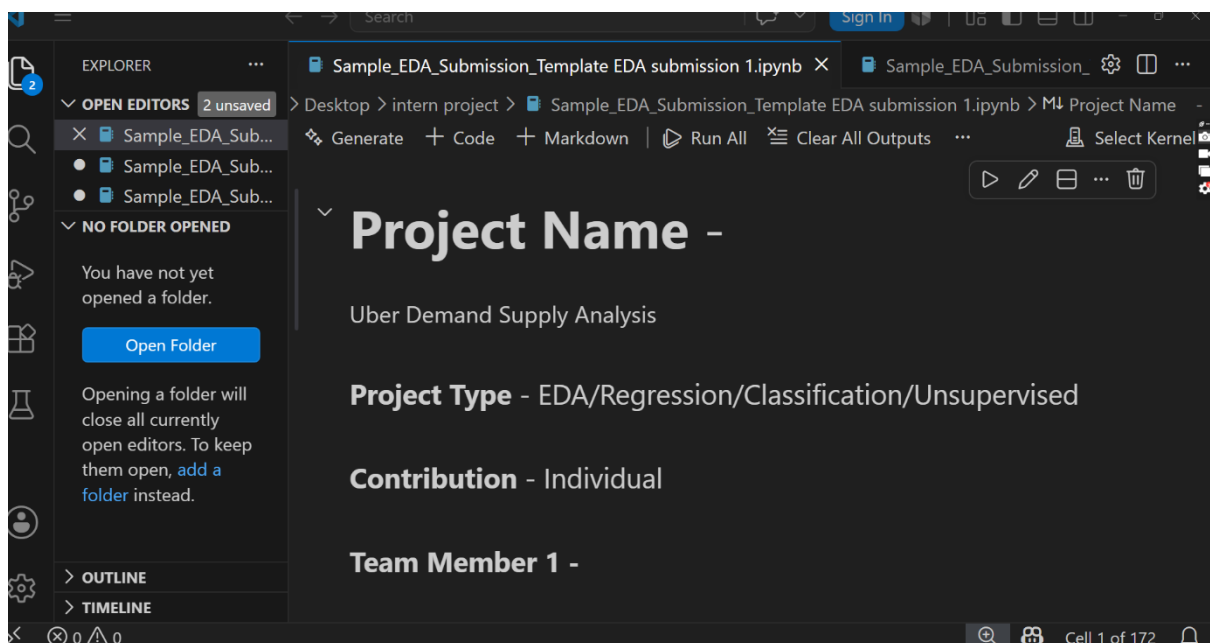
Recommendations

1. Increase driver availability during peak demand hours.
2. Provide incentives for drivers operating near airports.
3. Implement demand forecasting techniques for better driver allocation.
4. Monitor cancellation rates regularly.
5. Use real-time dashboards for operational decision-making.

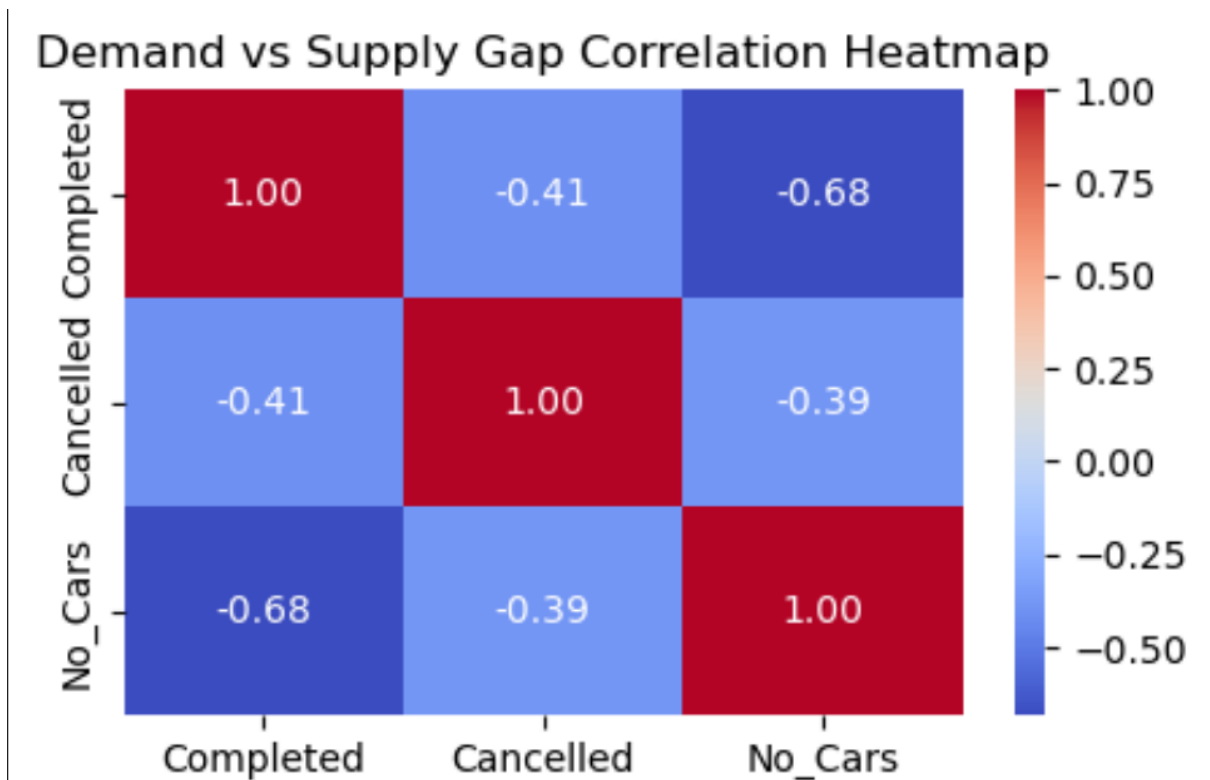
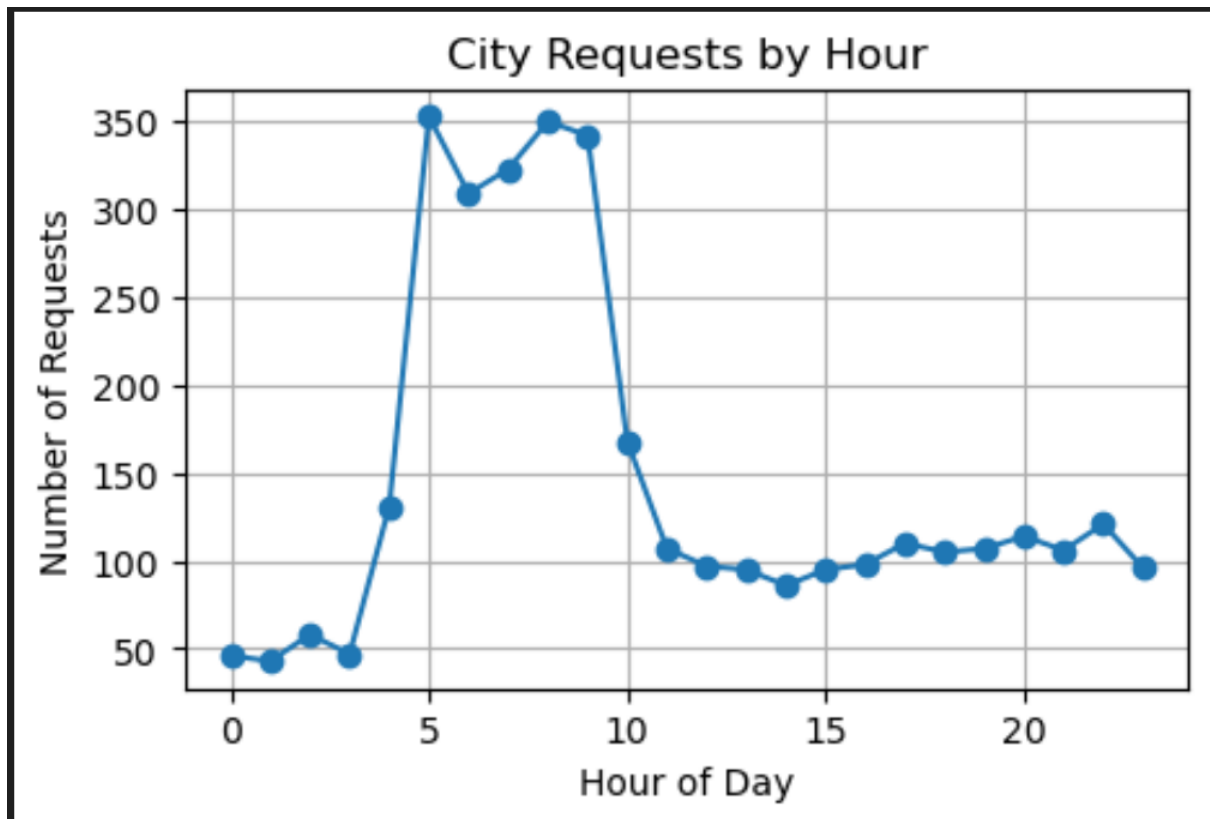
Conclusion

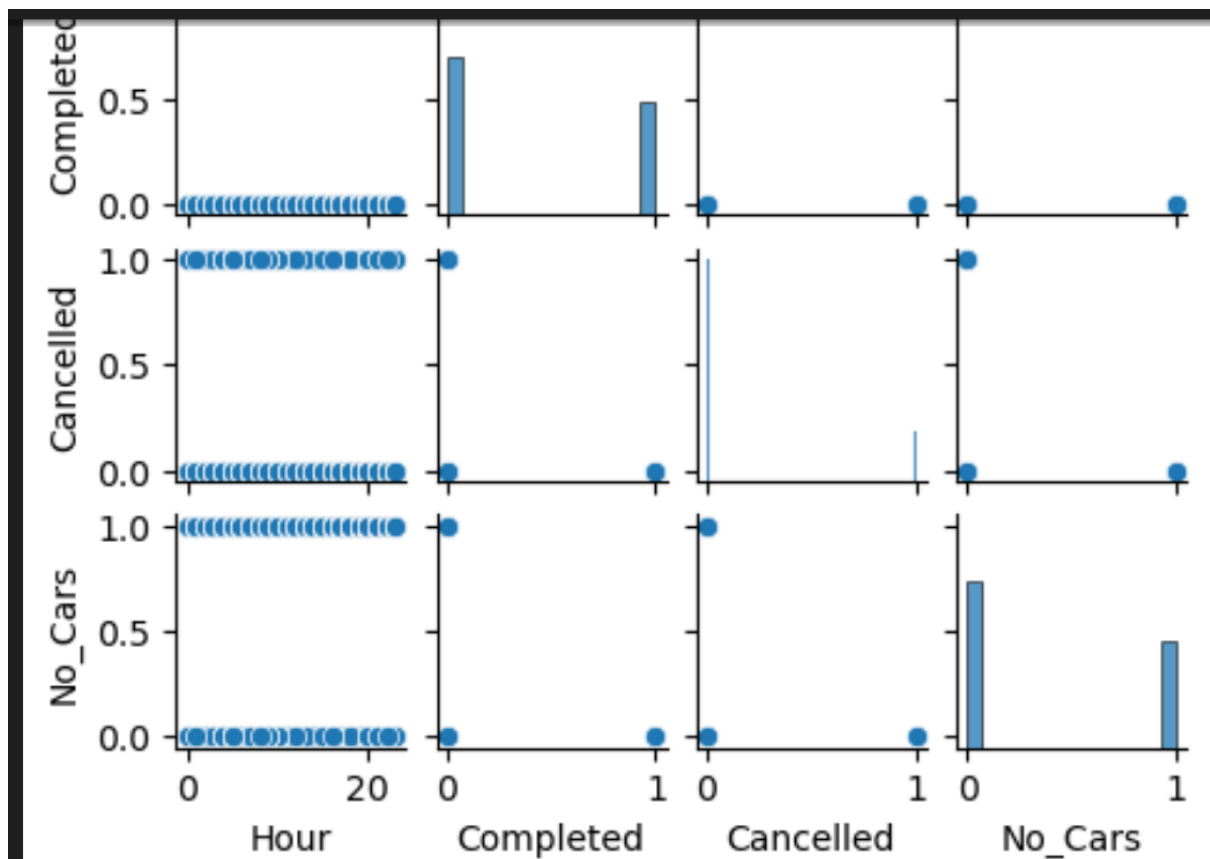
The Power BI dashboard successfully visualized key business metrics related to Uber's demand-supply gap. The analysis identified peak demand periods, driver availability issues, and location-specific challenges. These insights can help improve operational efficiency, customer satisfaction, and resource utilization

Python Exploratory Data Analysis



(Data Cleaning and Preprocessing)





Work Performed

- Data Cleaning
- Missing Value Analysis
- Feature Engineering
- Statistical Analysis
- Data Visualization

Key Findings

- Morning and evening hours showed highest demand.
- Driver shortages caused request failures.
- Airport pickup points experienced major supply gaps.
- Cancellation rates increased during peak demand periods.

GitHub Repository

Project Link:

<https://github.com/pallavisoni1800/uber-demand-supply-gap-analysis>

Conclusion

The project successfully identified the major causes of Uber's demand-supply gap. Using Excel, SQL, Python, and Power BI, important insights were generated regarding trip completion, cancellations, and driver availability. The recommendations can help improve customer satisfaction, optimize resources, and increase operational efficiency.